IMEN574 / IDSC574 · PROGRAMMING FOR AX

9

Classification

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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Classification

01 **Probability Models**

Fit scores and probabilities

02 **Decision Rules**

Trade misses against false alarms

03 **Reliable Evaluation**

Check probabilities and populations

01 · Probability Models

01 **Probability Models**

Fit scores and probabilities

02 **Decision Rules**

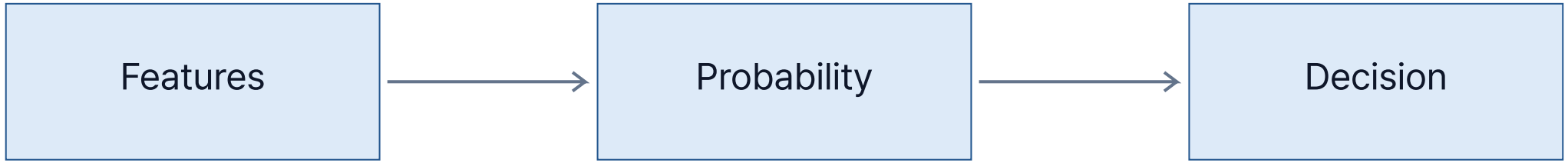
Trade misses against false alarms

03 **Reliable Evaluation**

Check probabilities and populations

Classification Workflow

A score becomes an action only after a decision rule



The model and decision rule solve different problems

Binary and Multiclass Targets

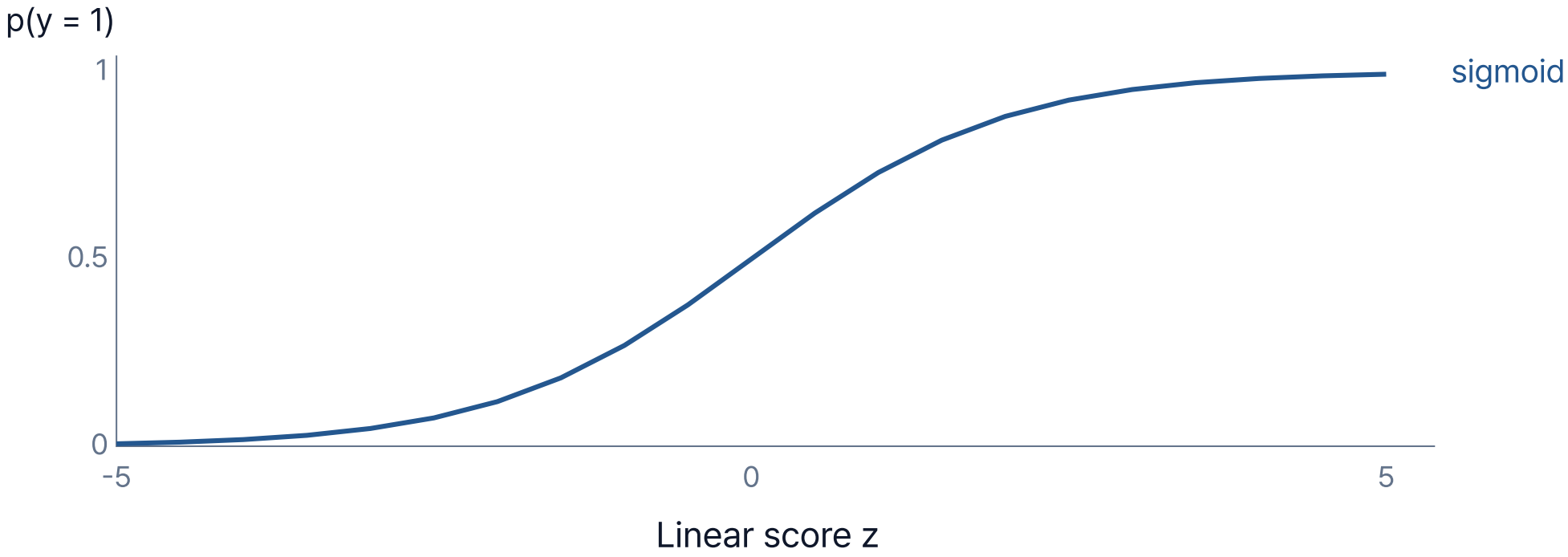
The target defines the output space

	y	Meaning
Label 0	0	normal
Label 1	1	fault

A binary label records one chosen event

Logistic Regression

A linear score is mapped into a probability

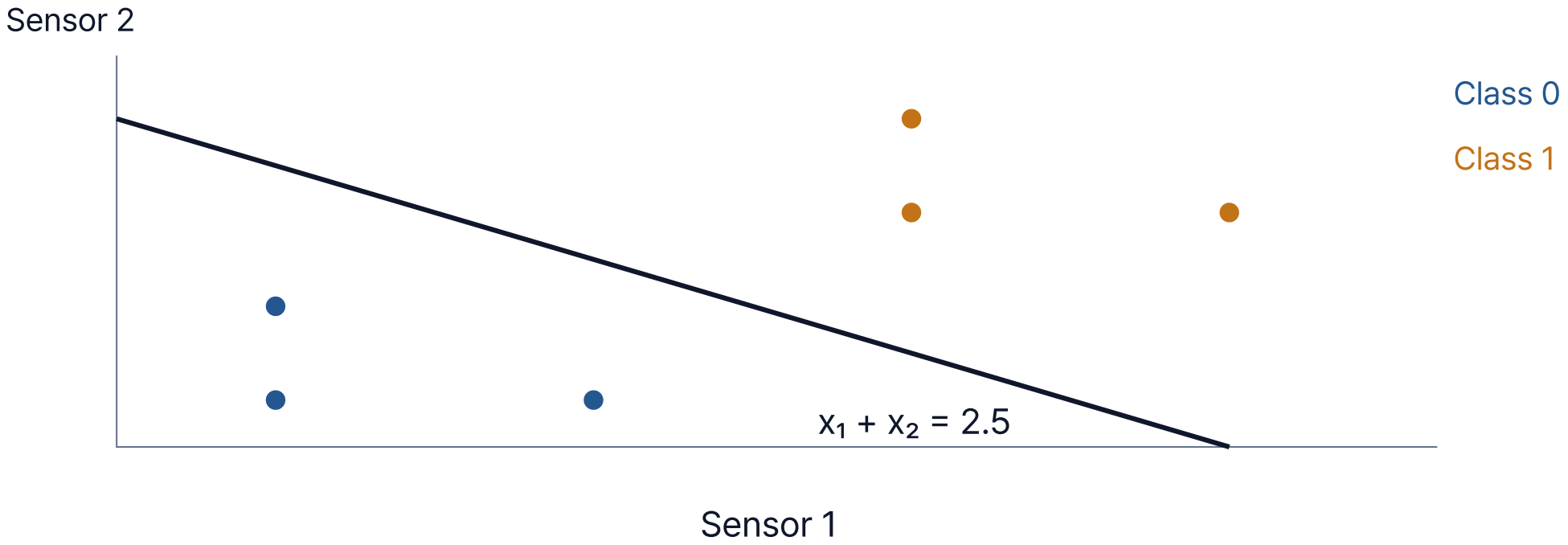


$$p = 1 / (1 + \exp(-z)), \; z = w^{T}x + b$$

Large scores approach one without exceeding it

Decision Boundary

With w = (1, 1) and b = −2.5, the boundary has $x_1 + x_2 = 2.5$

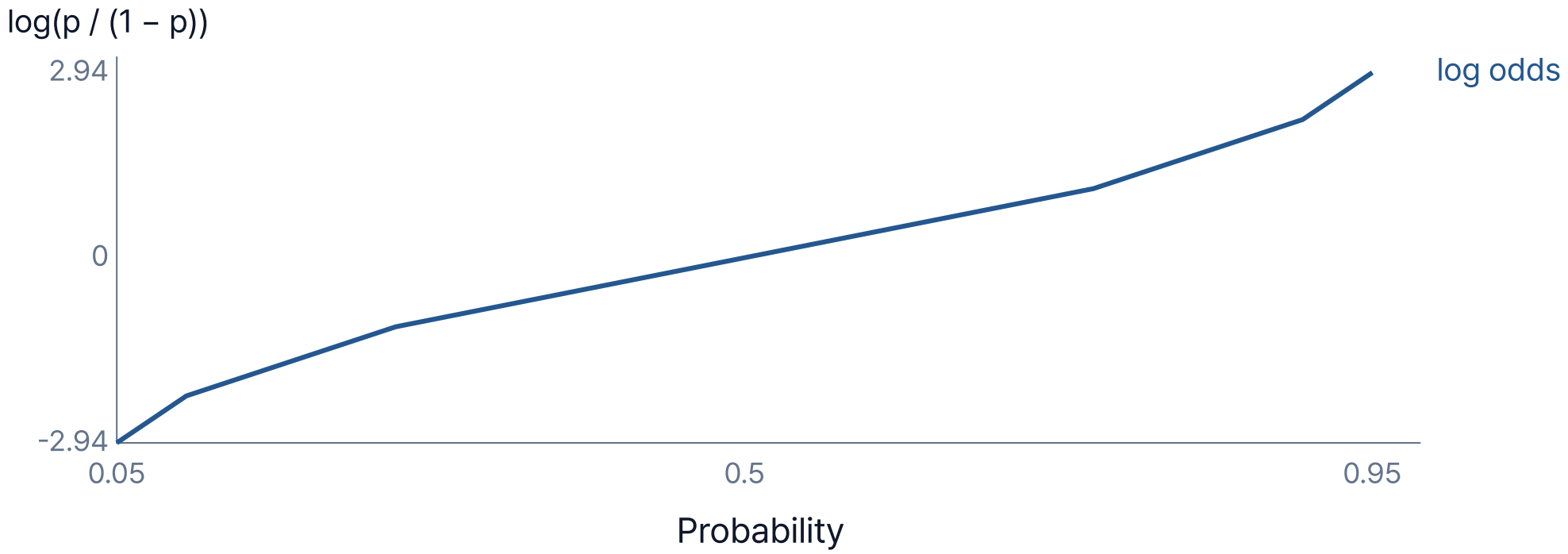


$$w^T x + b = 0 \text{ at } p = 0.5$$

A threshold selects one side of the score surface

Log Odds

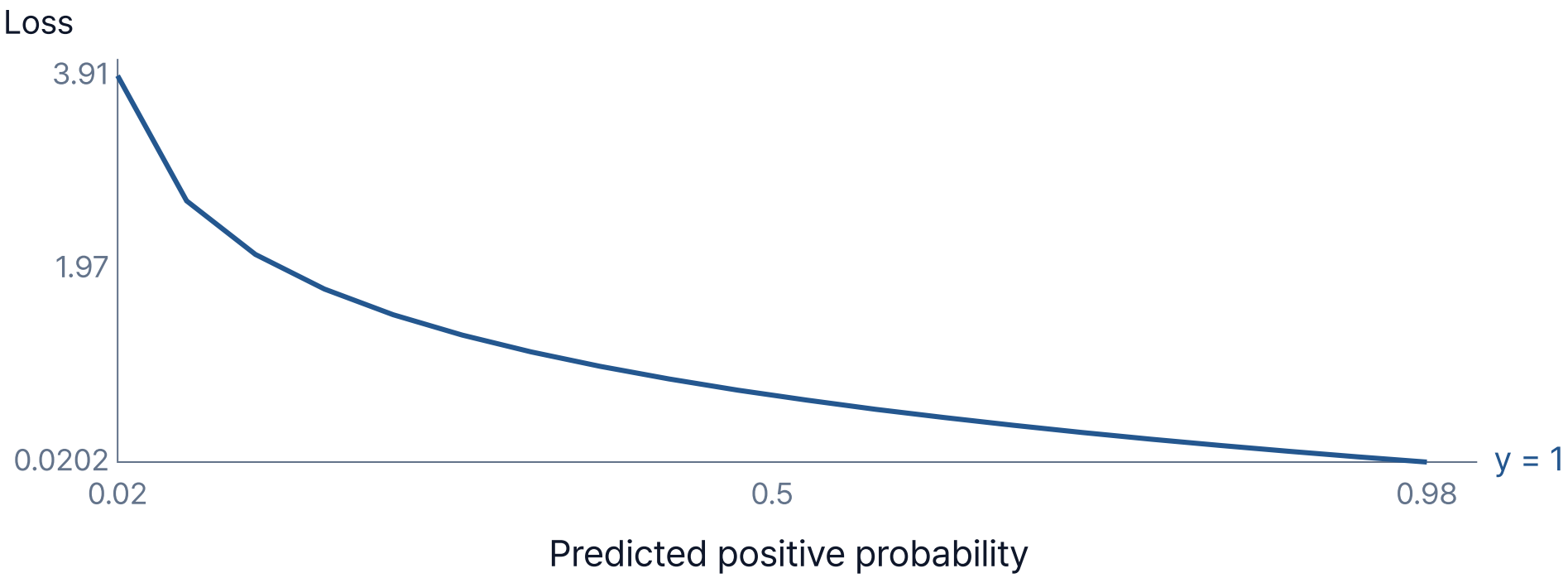
A coefficient changes odds multiplicatively



Probability changes are nonlinear even with linear log odds

Cross-Entropy Loss

Confident mistakes receive a large penalty



A positive example predicted near zero is costly

Stable Loss Calculation

Compute the loss from logits without taking log of zero

```
z = np.array([-4., 0., 4.])
y = np.array([0., 1., 1.])
loss = np.logaddexp(0, z) - y*z
print(np.round(loss, 4))
```

```
[0.0181 0.6931 0.0181]
```

Probability Gradient

The residual p − y drives each update

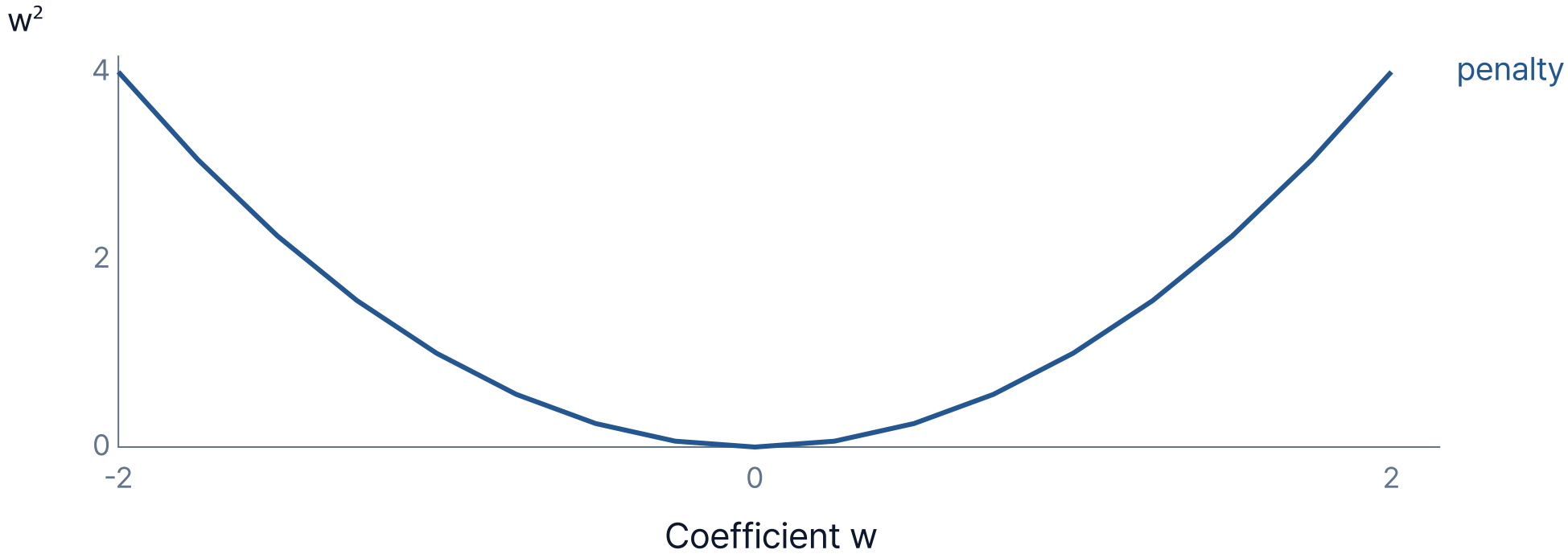
	y	p	p − y
Negative	0	0.2	0.2
Positive	1	0.2	-0.8

$$\partial L/\partial w = (p - y)x$$

The same prediction yields opposite correction directions

Regularized Classification

Penalize large coefficients to limit sensitivity



$$\text{Objective} = \text{mean cross-entropy} + \lambda \|w\|^2$$

The penalty grows with coefficient magnitude

Multiclass Softmax

Class probabilities share one normalization

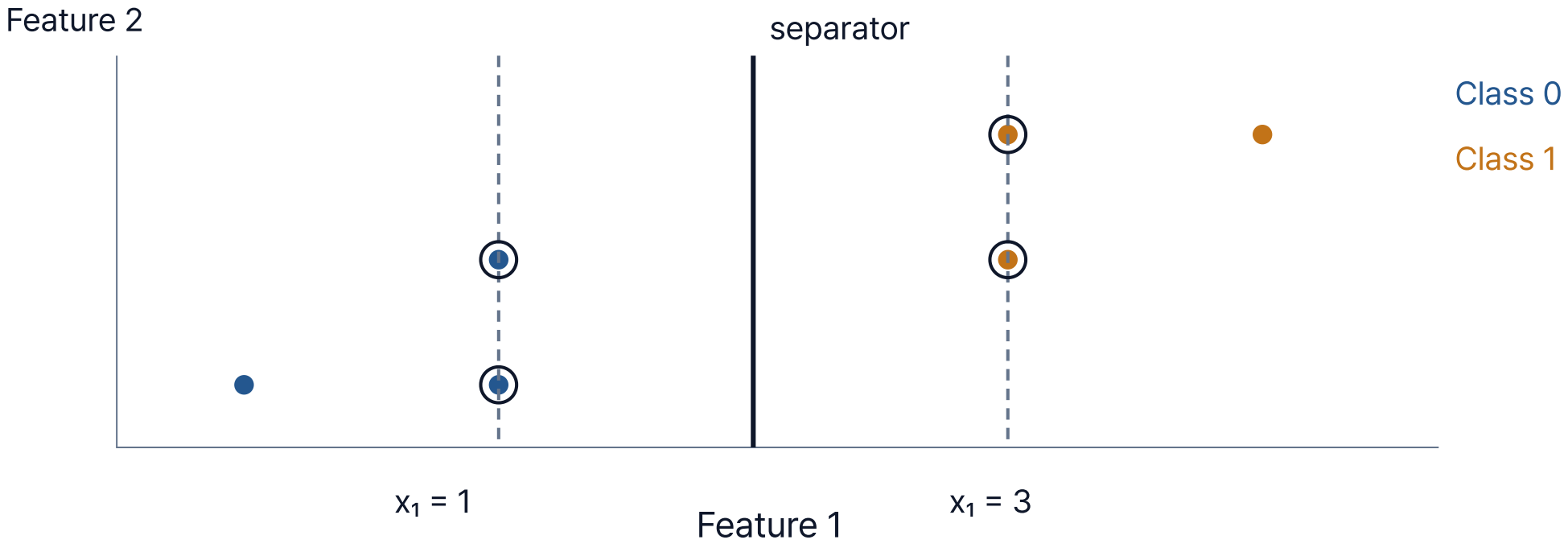
Class	Logit	Probability
A	2.0	0.665
B	1.0	0.245
C	0.0	0.09

$$p_k = \exp(z_k) / \Sigma_j \exp(z_j)$$

Increasing one logit lowers the relative share of others

Support Vector Classification

A margin separates fitting from distance to the boundary



Margin depends on the nearest training examples

The nearest points constrain the separating boundary

02 · Decision Rules

01 Probability Models

Fit scores and probabilities

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Predictions at a Threshold

One score list supports many decisions

	1	2	3	4	5	6
True y	0	0	1	0	1	1
Score	0.05	0.25	0.40	0.55	0.75	0.90
p ≥ 0.5	0	0	0	1	1	1

The third example is a missed positive at 0.5

Confusion Matrix

Count errors by their consequence

	Pred 0	Pred 1
True 0	2	1
True 1	1	2

One false alarm and one miss remain

Precision and Recall

Ask two different questions about positive cases



Ratio

Both equal two thirds on these six examples

F1 Score

The harmonic mean penalizes an unbalanced pair

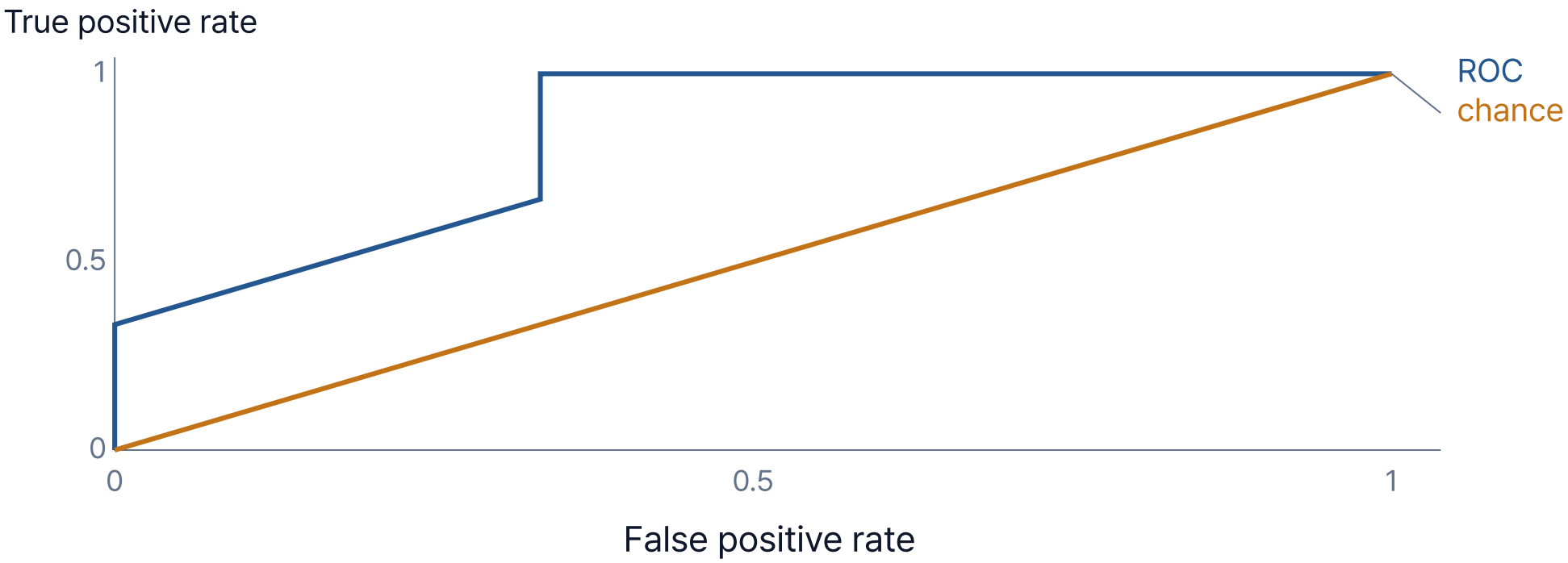
Precision	Recall	F1
1	0.25	0.4
0.5	0.5	0.5
0.25	1	0.4

$$F1 = 2PR / (P + R)$$

Perfect recall alone does not produce a high F1

ROC Space

Compare sensitivity with false-alarm rate



Thresholds trace a path through operating points

Precision-Recall Space

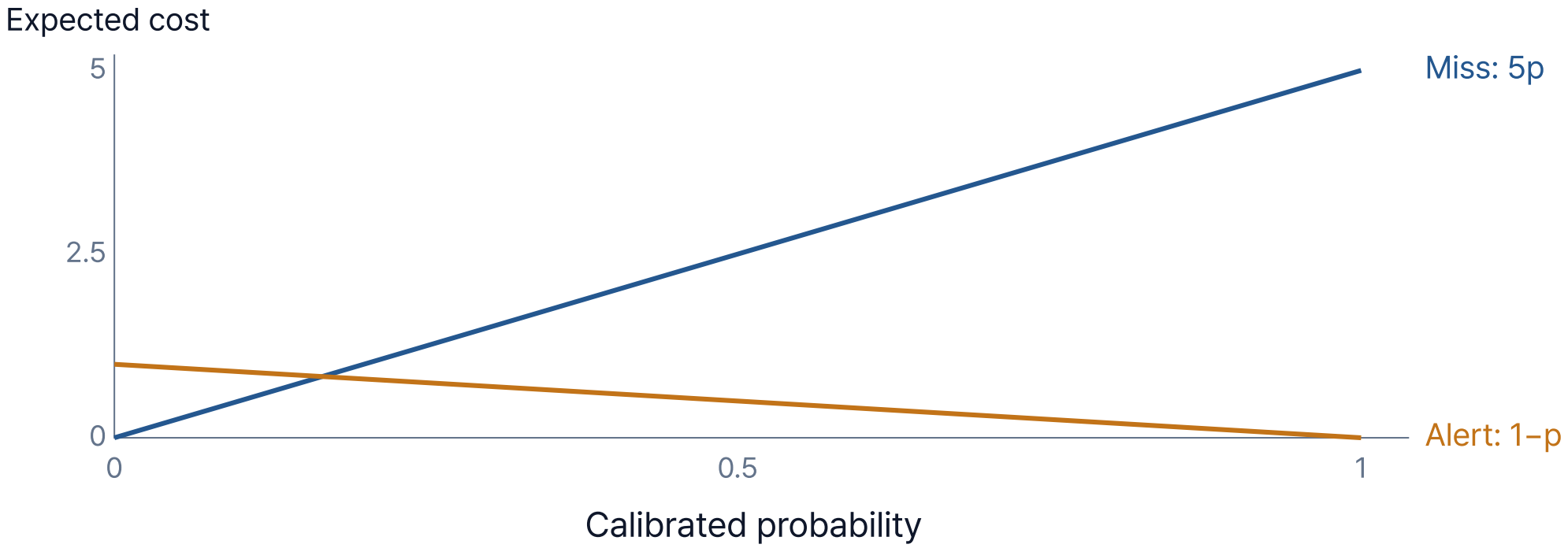
Positive prevalence changes the meaning of retrieved cases

TP	FP	Precision
10	90	0.1
10	10	0.5
10	0	1

The same ten true alerts can require very different review effort

Cost-Sensitive Decisions

Choose the action with lower expected cost

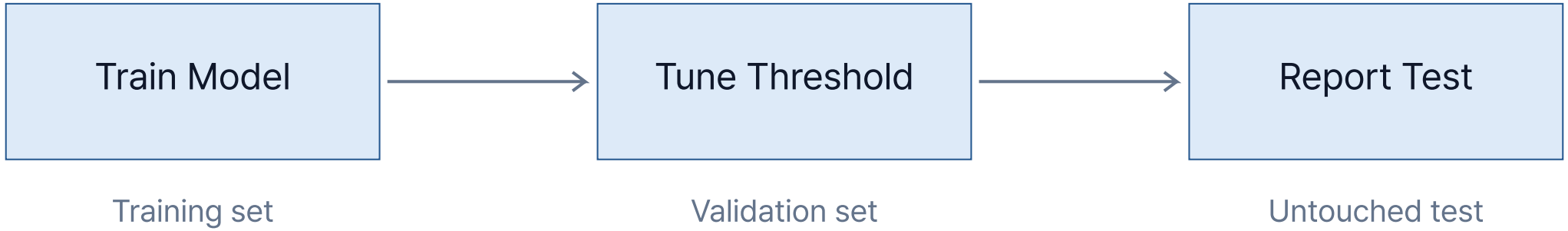



```
Alert when p > C_FP / (C_FP + C_FN)
```

Higher miss cost lowers the action threshold

Threshold Selection Split

Tune a decision rule on validation data



A threshold chosen on the test labels contaminates evaluation

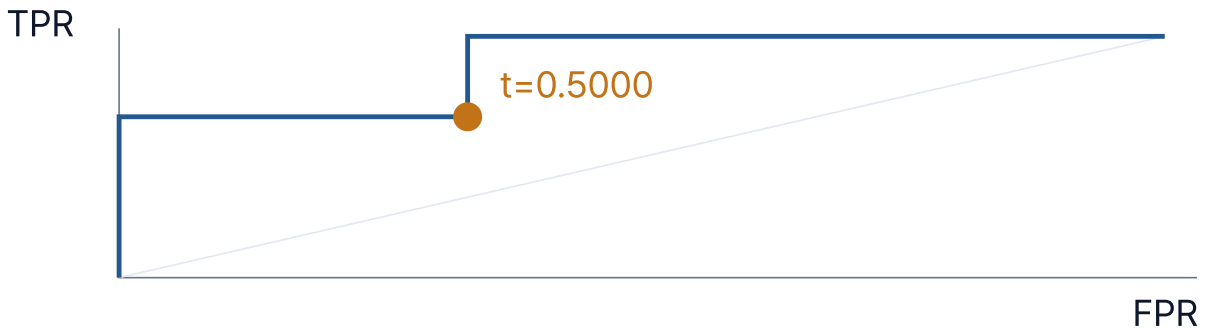
Demo: Threshold, ROC, and Imbalance

Which errors change at t = 0.3? Does repeating negatives change ROC or precision?

Score, label

```
0.05,0; 0.25,0; 0.40,1; 0.55,0;
0.75,1; 0.90,1
```

Threshold 0.5 Negative copies 1



TP 2, FP 1, FN 1, TN 2 | AUC 0.8889
Precision 0.6667, recall 0.6667, accuracy 0.6667
Cost = FP + 5FN = 6; negative rows repeated 1×

Exact counts from editable scores and labels

Answer: Threshold and Ranking

The same six scores give different actions at two thresholds

Threshold	TP / FP / FN / TN	Precision	Recall
0.5	2 / 1 / 1 / 2	0.667	0.667
0.3	3 / 1 / 0 / 2	0.750	1.000

Ranking stays fixed while the decision changes

Caveat: Rare Positive Decisions

Replicating negatives increases review burden without changing this ROC

Negative Copies	TP	FP	Precision
1	2	1	2/3
5	2	5	2/7

The same ranking can produce many more false alarms

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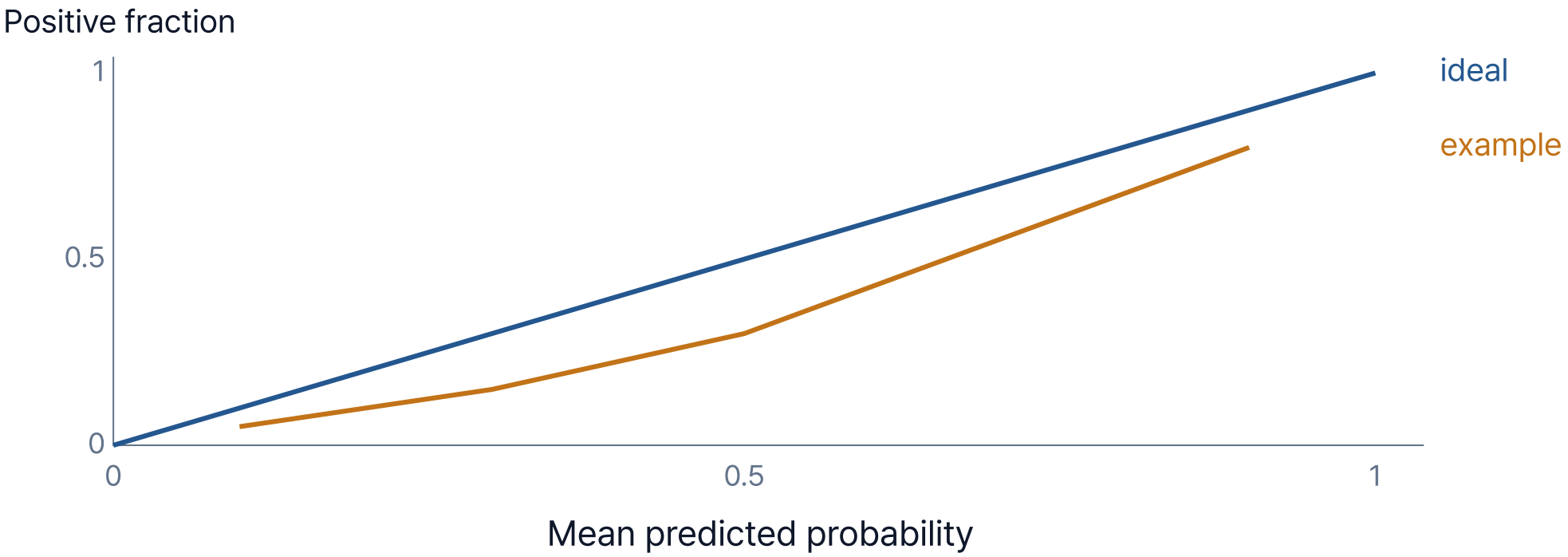
Trade misses against false alarms

03 **Reliable Evaluation**

Check probabilities and populations

Probability Calibration

Predicted probabilities should match observed frequencies



This example predicts more positives than each bin observes

Brier Score

Measure squared probability errors rather than only labels

y	p	$(p - y)^2$
1	0.9	0.01
1	0.6	0.16
0	0.6	0.36

The score distinguishes predictions with the same hard label

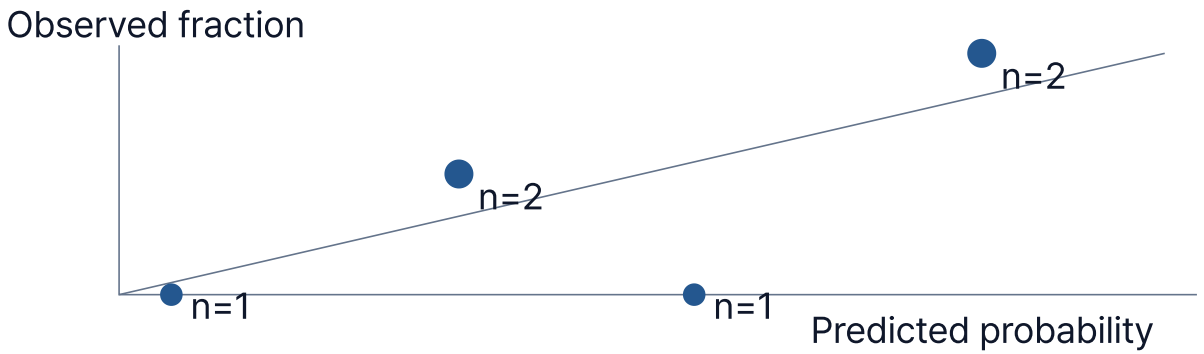
Demo: Probability Calibration Arithmetic

Does sharpening improve Brier score? Do six rows establish reliable calibration?

Probability, label

```
0.05,0; 0.25,0; 0.40,1; 0.55,0; 0.75,1; 0.90,1
```

Logit temperature 1



T 1 | Brier 0.1333 | log loss 0.4078
First 6 / 6 probabilities: 0.05000, 0.2500, 0.4000, 0.5500, 0.7500, 0.9000
Four fixed bins; six examples illustrate arithmetic, not a calibration guarantee

Toy reliability bins; no calibration model is fitted

Answer: Probability Error

Compute probability losses from all six held-out labels

T	Brier	Log Loss
1.0	0.133333	0.407803
0.5	0.143043	0.386372

A sharper probability is useful only when its evidence supports it

Caveat: Tiny Reliability Bins

A few observed cases cannot establish a stable calibration curve

Bin	Rows	Positive Fraction
[0,0.25)	1	0
[0.25,0.5)	2	0.5
[0.5,0.75)	1	0
[0.75,1]	2	1

Small bin counts explain the irregular observed fractions

Uncertainty Decomposition

Confidence can fail for different reasons

Source	Example	Response
Data noise	Ambiguous label	Improve measurement
Limited coverage	New machine	Collect new evidence
Model mismatch	Linear boundary	Compare alternatives

A confident score does not identify its source of error

Stratified Validation

Preserve rare labels when estimating ordinary random splits

	Class 0	Class 1
Fold A	18	2
Fold B	18	2
Fold C	18	2

Each example fold contains both classes

Group-Aware Evaluation

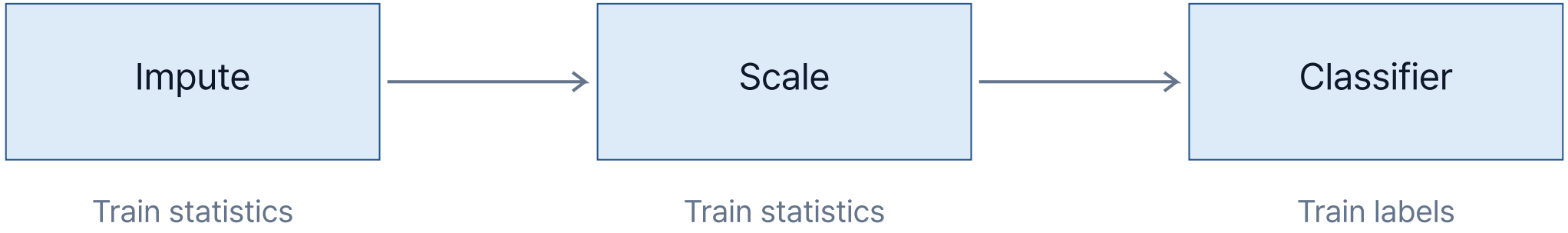
Keep examples from one entity on one side

	Entity 1	Entity 2	Entity 3
Train	A	A	B
Test	C	C	D

Repeated measurements require an entity boundary

Classification Pipeline

Fit every learned transformation inside the training split



Validation rows pass through fixed transformations

Auditing Generated Predictions

Verify the positive label and metric inputs explicitly

```
y = np.array([0, 0, 1, 0, 1, 1])
p = np.array([.05, .25, .4, .55, .75, .9])
a = p >= .5
print("TP", int(((y==1)&a).sum()))
print("FP", int(((y==0)&a).sum()))
print("FN", int(((y==1)&~a).sum()))
```

```
TP 2
FP 1
FN 1
```

Class Weighting

Changing the loss changes what fitting emphasizes

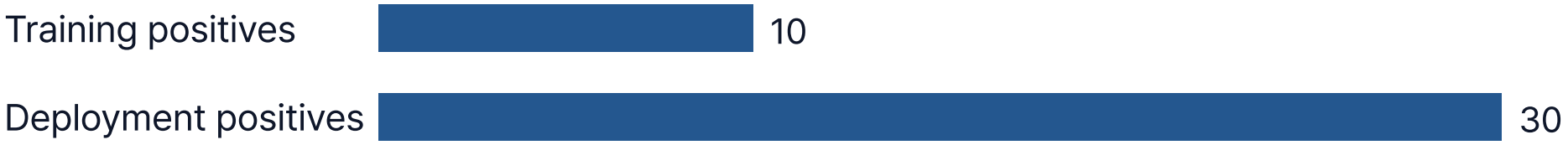


Loss multiplier

A weighted positive error contributes five times as much

Distribution Shift

A fixed rule can meet a different positive rate



Cases per 100

The event becomes more common in the second population

Model Selection Record

Retain the assumptions behind the selected operating point

Item	Recorded Choice
Positive class	Fault requiring inspection
Split unit	Machine
Primary metric	Recall at review budget
Threshold	Chosen on validation
Final report	Frozen test once

An operating point is meaningful only with its evaluation protocol

Takeaway: Classification

01 Separate probability estimation from action selection

02 Tune thresholds using validation data

03 Audit rare events and deployment shifts